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Option Hedging under Volatility Risk

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Introduction

Risk management in options trading has long been a critical concern in financial markets. The standard reference model, developed by Black, Scholes, and Merton, relies on several strong assumptions: absence of arbitrage, perfect market liquidity, continuous trading, and most notably, constant volatility. While the latter assumption simplifies analytical computations, it significantly departs from observed market behavior, where volatility is dynamic, stochastic, and often unpredictable. The major volatility shocks during the 2008 financial crisis and, more recently, the COVID-19 pandemic, have clearly illustrated the practical limitations of this classical model.

Against this backdrop, a fundamental question arises: *Can we rely solely on a dynamic hedging strategy based on delta adjustments, when the main source of uncertainty often lies in unexpected volatility changes?* More precisely, to what extent does variable volatility impact the performance of a traditional delta-hedging strategy for vanilla option portfolios? This dissertation seeks to address that very issue.

To do so, we conduct a rigorous comparative analysis across three distinct modeling frameworks:

- **The Black-Scholes model**, with constant volatility, serving as the theoretical benchmark;
- **The Heston model**, in which volatility follows a continuous-time stochastic process, allowing for the replication of empirically observed phenomena such as volatility smiles, skews, and asset-volatility correlation;
- **The GARCH(1,1) model**, drawn from econometric literature, in which conditional variance is modeled from historical data, offering a more empirical perspective on market risk.

Other models, such as the **Dupire local volatility model** or the **Hull-White model** with stochastic interest rates, are also used in practice to capture volatility surfaces or rate dynamics. However, these models fall outside the scope of this dissertation, which focuses exclusively on constant, stochastic, and data-driven volatility models.

These three frameworks will allow us to assess how explicitly incorporating volatility dynamics affects the effectiveness of dynamic hedging strategies. We will implement several strategies: the standard delta-only hedge, and more advanced approaches involving gamma and vega neutralization. In this context, we will also evaluate the practical relevance of volatility hedging instruments such as Variance Swaps and VIX Futures.

Our methodological approach follows a clearly structured sequence. First, a comprehensive theoretical review of the selected models will be presented. Next, we introduce the dynamic hedging strategies and instruments available to hedge volatility risk. Finally, we perform an in-depth numerical analysis based on market simulations under each model.

This comparative process will enable us to quantify the true impact of volatility dynamics on the performance of various hedging strategies.

This dissertation is organized as follows. After this general introduction, **Chapter 1** provides a detailed review of the selected volatility models (Black-Scholes, Heston, and GARCH). **Chapter 2** presents the dynamic hedging strategies commonly used in financial markets. **Chapter 3** focuses on instruments specifically designed to hedge volatility risk. **Chapter 4** outlines the simulation framework and provides a comparative analysis of the results. **Chapter 5** offers a critical discussion of the findings, before concluding with the key takeaways and potential avenues for further research.

Chapter 1
Theoretical Review of Volatility Models

1.1 The Black–Scholes Model (1973)

The Black–Scholes model, introduced by Fischer Black and Myron Scholes and extended by Robert Merton, remains the canonical framework for pricing European options and managing the associated risks. It is built on a set of fundamental assumptions that render the model analytically tractable and serve as a benchmark for more sophisticated extensions.

1. Fundamental Assumptions

- **Absence of arbitrage:** Markets do not allow a riskless profit with zero investment.
- **Frictionless markets:** No transaction costs or taxes; assets are perfectly divisible; borrowing and lending occur at the same risk-free rate.
- **Constant, known volatility:** The volatility parameter σ, representing the typical amplitude of underlying-price fluctuations, is assumed constant and known.
- **Continuous trading:** The underlying price evolves continuously in time and is modeled by a continuous-time stochastic process (geometric Brownian motion).
- **Constant interest rate:** The risk-free rate r remains constant over the life of the option.

2. Introduction to Brownian Motion

Existence theorem: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. There exists a stochastic process $(B_t)_{t\geq 0}$ with continuous paths (almost surely) such that $B_0 = 0$ and, for all $0 \leq s \leq t$:

$$\begin{cases} B_t - B_s \sim \mathcal{N}(0,\, t-s), \\ B_t - B_s \text{ is independent of } \sigma\Big(B_u : u \leq s\Big). \end{cases}$$

In particular, $B_t \sim \mathcal{N}(0, t)$.

3. Underlying Asset Dynamics

Under these assumptions, the underlying asset price S_t follows a geometric Brownian motion:

$$\boxed{S_t = \mu S_t \, t \; + \; \sigma S_t \, W_t,}$$

where

- S_t is the asset price at time t,
- μ is the instantaneous drift,
- σ is the (constant) volatility,
- W_t is a standard Brownian motion.

Proof of Existence (Log-Normality)

Assume $\log S_t$ is normal, so that

$$\log S_t = \log S_0 + \left(\mu - \tfrac{1}{2}\,\sigma^2\right) t + \sigma \, W_t.$$

Exponentiating yields the geometric Brownian motion:

$$S_t = S_0 \exp\Big[\Big(\mu - \tfrac{1}{2}\,\sigma^2\Big)t + \sigma \, W_t\Big].$$

In discrete time, by discretizing the continuous-time solution over a single time step Δt and setting

$$W_{t+\Delta t} - W_t \;=\; \sqrt{\Delta t}\, Z, \quad Z \sim \mathcal{N}(0,1),$$

Hence

$$S_{t+\Delta t} = S_t \exp\Big((\mu - \tfrac{1}{2}\sigma^2)\, \Delta t + \sigma \, \sqrt{\Delta t} \; Z\Big).$$

4. Application of Itô's Lemma

Define

$$f(t, W_t) \;=\; S_0 \exp\Big[\Big(\mu - \tfrac{1}{2}\,\sigma^2\Big)t + \sigma \, W_t\Big].$$

By Itô's lemma,

$$S_t = \left(\partial_t f + \tfrac{1}{2}\,\partial^2_{W_t} f\right) dt + \partial_{W_t} f \, dW_t.$$

Since

$$\partial_t f = \left(\mu - \tfrac{1}{2}\sigma^2\right) S_t, \quad \partial_{W_t} f = \sigma S_t, \quad \partial^2_{W_t} f = \sigma^2 S_t,$$

we recover

$$dS_t = S_t\Big[(\mu - \tfrac{1}{2}\sigma^2) + \tfrac{1}{2}\sigma^2\Big]\, dt + \sigma S_t \, dW_t = \mu S_t \, dt + \sigma S_t \, dW_t.$$

5. Black–Scholes Formula for European Calls

Under no-arbitrage and risk-neutral valuation, the price at $t = 0$ of a European call with strike K and maturity T is

$$\boxed{C(S_0, 0) \;=\; S_0\, N(d_1) \;-\; Ke^{-rT}\, N(d_2),}$$

where

$$d_1 = \frac{\ln\!\big(S_0/K\big) + (r + \tfrac{1}{2}\,\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T},$$

and $N(\cdot)$ is the standard normal CDF. An analogous formula holds for puts via put–call parity.

6. Delta, Gamma and Vega

In the Black–Scholes framework, it is essential to understand how an option's value responds to small changes in market parameters. These sensitivities—known as the "Greeks"—are critical for effective risk management of option portfolios:

- **Delta** measures the sensitivity of the option price to an infinitesimal change in the underlying asset price. Formally, for a call option:

  $$\Delta = \frac{\partial C}{\partial S}.$$

  This quantity approximates the change in the option price for a small move in the underlying.

- **Gamma** measures the sensitivity of Delta itself to a change in the underlying price, i.e. the second derivative of the option price with respect to the underlying:

  $$\Gamma = \frac{\partial^2 C}{\partial S^2}.$$

- **Vega** measures the sensitivity of the option price to a change in the implied volatility σ:

  $$\text{Vega} = \frac{\partial C}{\partial \sigma}.$$

Mastering these sensitivities is crucial for constructing and maintaining hedges that minimize the risk of an option portfolio.

7. Perfect Delta Hedging in Black–Scholes

To replicate the option dynamically, consider a portfolio

$$\Pi_t = \Delta_t S_t \; + \; \varphi_t B_t,$$

where $B_t = e^{rt}$ is the risk-free asset. One chooses

$$\Delta_t = \partial_S V(S_t, t), \qquad \varphi_t = \frac{V(S_t, t) - \Delta_t S_t}{B_t},$$

so that $\Pi_t = V(S_t, t)$ and the stochastic parts match exactly. This yields the Black–Scholes PDE:

$$\partial_t V + \tfrac{1}{2}\,\sigma^2 S^2 \partial_{SS}^2 V + rS\,\partial_S V - rV = 0.$$

8. Limitations of the Black–Scholes Model

The Black–Scholes model, although elegant and foundational in option pricing theory, presents several limitations that restrict its applicability in real-world financial markets:

- **Constant volatility:** The model assumes a constant volatility σ, which does not reflect market reality. In practice, volatility changes over time and varies with the level of the underlying asset, as evidenced by the "volatility smile" below.

- **Log-normal return distribution:** Returns are assumed to follow a normal distribution, whereas empirical market data exhibit heavy tails (high kurtosis) that deviate significantly from normality.

- **Continuous hedging:** The model requires continuous rebalancing of the replicating portfolio without transaction costs. In reality, hedges are adjusted discretely and incur bid–ask spreads, commissions, and slippage.

- **No price jumps:** Sudden market shocks (e.g. news events or crises) are not captured by the continuous diffusion assumption, making the model unreliable during turbulent periods.

- **No early exercise or dividends:** The standard formula applies only to European options on non-dividend-paying assets, limiting its relevance for American-style contracts or stocks with discrete dividend payments.

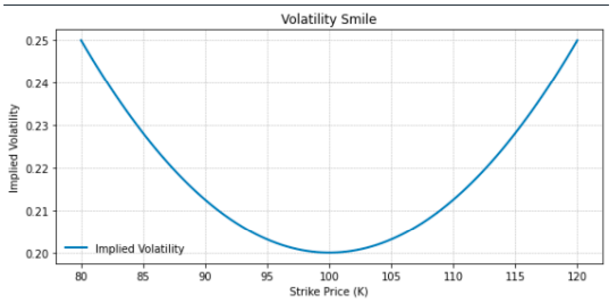


Figure 1: Volatility smile: implied volatility varies with strike price, contradicting the Black–Scholes assumption of constant volatility.

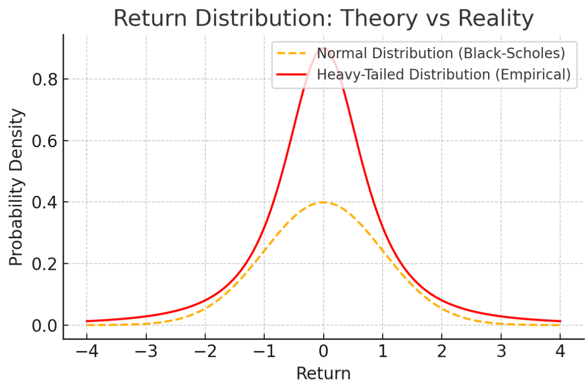


Figure 2: Return distribution: Black-Scholes assumes normality, but real markets show heavy tails (more frequent extreme events).

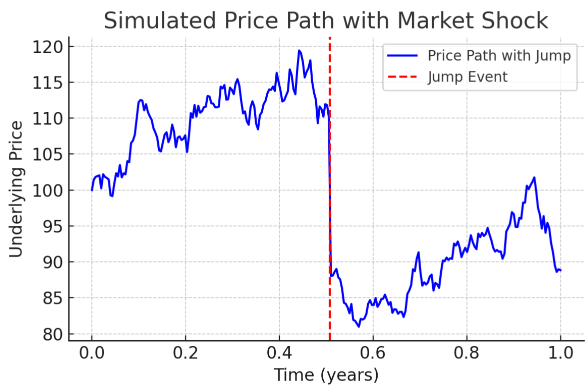


Figure 3: Simulated price path with a sudden drop: Black–Scholes cannot capture such market shocks.

1.2 Heston Model: Equations, Interpretation and Implementation

1. Model Equations

1.1 SDE under the Risk-Neutral Measure:

$$\begin{cases} dS_t = (r - q)\, S_t\, dt \;+\; \sqrt{v_t}\, S_t\, dW_t^S, \\ dv_t = \kappa\, (\theta - v_t)\, dt \;+\; \sigma_v\, \sqrt{v_t}\, dW_t^v, \end{cases}$$

- r is the risk-free rate
- q is the dividend yield
- The Brownian increments satisfy: $\langle dW_t^S,\, dW_t^v \rangle = \rho\, dt, \quad |\rho| \leq 1$

1.2 Alternative Form (Orthogonal Decomposition)

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - q)\, dt + \sqrt{v_t}\left(\rho\, dz_t^v + \sqrt{1 - \rho^2}\, dw_t\right), \\ dv_t &= \kappa\, (\theta - v_t)\, dt + \sigma_v\, \sqrt{v_t}\, dz_t^v. \end{aligned}$$

where dz_t^v and dw_t are independent, so that the uncorrelated component of volatility is captured by dw_t.

2. Parameter Interpretation

The Heston model involves five intuitive parameters:

- v_0: initial variance at $t = 0$.
- θ: long-term mean variance toward which v_t reverts.
- κ: speed of mean reversion; higher κ faster pull to θ.
- σ_v: "vol-of-vol," controlling the magnitude of variance fluctuations.
- ρ: correlation between spot and variance shocks; $\rho < 0$ generates the equity skew.

Feller condition:

$$2\, \kappa\, \theta \;\geq\; \sigma_v^2,$$

which ensures that $v_t > 0$ almost surely.

3. Intuition on Negative Correlation and Skew

- **Impact of $\rho < 0$:** When the underlying falls ($dW_t^S < 0$), the correlated variance shock dW_t^v tends to be positive, boosting v_t and lifting out-of-the-money puts.
- **Skew Formation:** Lower strikes exhibit higher implied vol because large downward moves in S_t often coincide with volatility spikes.
- **Practical Consequence:** Heston can match both the level and slope (skew) of the market volatility surface, unlike Black–Scholes.

4. Key Sensitivities (Greeks)

In the Heston framework, the main sensitivities can be summarized in a highly compact form:

$$\Delta = P_1.$$

$$\frac{\partial C}{\partial v_0} = S_0\, D_1 \;-\; K\, e^{-rT}\, D_2.$$

All other Greeks (Gamma, Vega, Theta, Rho, …) are obtained by differentiating directly, under the Fourier-integral sign, the functions $C(u,t)$ and $D(u,t)$ that enter the Heston characteristic function.

1.3 Garch(1,1) Model: Empirical Volatility Dynamics

1. Discrete-Time Variance Recursion:

Let $\{u_n\}_{n\geq 1}$ denote the demeaned daily log-returns, and let $\mathcal{F}_{n-1} = \sigma(u_{n-1}, u_{n-2}, \dots)$ denote the available information.

The conditional variance follows the recursion:

$$\boxed{\sigma_{n+1}^2 = \omega + \alpha u_n^2 + \beta \sigma_n^2, \quad \omega > 0,\ \alpha, \beta \geq 0.} \tag{G11}$$

with the process:

$$u_n = \sigma_n \varepsilon_n, \quad \varepsilon_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1).$$

2. Stationarity and Variance:

The process is strictly stationary if and only if:

$$\alpha + \beta < 1.$$

Then the variance is:

$$\mathbb{E}[\sigma_n^2] = \frac{\omega}{1 - (\alpha + \beta)} := \bar{\sigma}^2.$$

3. Volatility Clustering:

The autocorrelation function of squared returns is:

$$\text{ACF}(k) = \frac{\text{Cov}(u_n^2, u_{n-k}^2)}{\text{Var}(u_n^2)} = (\alpha + \beta)^k, \quad k \geq 0.$$

This exponential decay captures the empirically observed phenomenon of *volatility persistence*.

4. Maximum-Likelihood Estimation:

Given a time series $\{u_1, \dots, u_N\}$, the log-likelihood function is:

$$\ell(\omega, \alpha, \beta) = -\frac{1}{2} \sum_{n=1}^{N} \left[\ln(2\pi) + \ln \sigma_n^2 + \frac{u_n^2}{\sigma_n^2} \right],$$

with σ_n^2 satisfying equation (G11).
Constraints:

- $\omega > 0$
- $\alpha, \beta \geq 0$
- $\alpha + \beta < 1$

5. Nature of the Model: Historical vs. Risk-Neutral

- GARCH is estimated under the **physical measure** $\mathbb{P}$, using historical returns.
- No-arbitrage is **not** guaranteed.
- For pricing, a measure change $\mathbb{P} \to \mathbb{Q}$ is required.

6. Pricing Adaptations:

Duan (1995) – Locally Risk-Neutral Valuation (LRNV). Introduces a market-price-of-risk parameter λ; option valuation via Monte Carlo simulation with importance sampling.

Heston–Nandi (2000). In order to price and hedge options in an arbitrage-free framework, we must move from the *physical* measure $\mathbb{P}$, under which we calibrate GARCH(1,1) on historical returns, to the *risk-neutral* measure $\mathbb{Q}$. Under $\mathbb{Q}$, the discounted underlying price must be a martingale:

$$\widetilde{S}_t = e^{-rt} S_t, \quad \mathbb{E}_{\mathbb{Q}}\left[\widetilde{S}_{t+\Delta t} \mid \mathcal{F}_t\right] = \widetilde{S}_t.$$

A direct *local risk-neutral* adjustment (Duan, 1995) shifts the innovation in the GARCH variance, but does not in general yield closed-form option prices.

Heston and Nandi (2000) propose instead a special GARCH(1,1) specification *already* under $\mathbb{Q}$:

$$\sigma_{n+1}^2 = \omega + \alpha\left(u_n - \gamma\sqrt{\sigma_n^2}\right)^2 + \beta\,\sigma_n^2, \quad \gamma = -\tfrac{1}{2},$$

The choice $\gamma = -\frac{1}{2}$ guarantees that the log-price is conditionally Gaussian under $\mathbb{Q}$, which leads to a closed-form characteristic function and fast Fourier-transform pricing of European options.

Interpretation:
By embedding the risk-premium directly into the variance recursion and enforcing the martingale property on S_t, the Heston–Nandi model aligns historical volatility dynamics with no-arbitrage pricing. In our implementation, we calibrate (ω, α, β) on $\mathbb{P}$ and then simulate under the Heston–Nandi law to obtain risk-neutral trajectories for delta-hedging.

7. Sensitivities and Long Memory:

Sensitivity of the unconditional variance:

$$\frac{\partial \bar{\sigma}^2}{\partial \alpha} = \frac{\omega}{[1-(\alpha+\beta)]^2}, \quad \frac{\partial \bar{\sigma}^2}{\partial \beta} = \frac{\omega}{[1-(\alpha+\beta)]^2}.$$

As $\alpha + \beta \to 1^-$, $\bar{\sigma}^2$ tends to infinity:

$$\Rightarrow \text{ fat tails and long-memory effects.}$$

Chapter 2
Dynamic Hedging Strategies

2.1 Delta Hedging:

1. Probabilistic Framework

We assume a frictionless market with a constant short rate r.

The underlying asset follows:

$$\boxed{dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}, \quad S_0 > 0,\ \sigma > 0} \tag{1}$$

The risk-free asset evolves as:

$$dB_t = rB_t\,dt \quad (B_0 = 1) \;\Rightarrow\; B_t = e^{rt}.$$

Let $V(S,t) \in C^{1,2}([0,T) \times \mathbb{R}_+)$ be the price of a European option.

2. Itô's Lemma Applied to the Option Price

Under the risk-neutral measure $\mathbb{Q}$:

$$\boxed{dV_t = \Big(\partial_t V + rS_t\partial_S V + \tfrac{1}{2}\sigma^2 S_t^2 \partial_{SS}^2 V\Big)\,dt + \sigma S_t \partial_S V\,dW_t^{\mathbb{Q}}} \tag{2}$$

The stochastic term (in dW_t) represents the risk exposure.

3. Self-Financing Replicating Portfolio

Definition:

$$\Pi_t = \Delta_t S_t + \varphi_t B_t$$

Self-financing condition:

$$d\Pi_t = \Delta_t\,dS_t + \varphi_t\,dB_t$$

Portfolio Dynamics. Using equation (1):

$$\boxed{d\Pi_t = (r\Delta_t S_t + r\varphi_t B_t)\,dt + \sigma\Delta_t S_t\,dW_t^{\mathbb{Q}}} \tag{3}$$

4. Delta Hedging Condition

We require:

$$\Pi_t = V_t \quad \text{and} \quad d\Pi_t = dV_t$$

(a) Matching the stochastic terms: From (2) and (3), compare the coefficients of $dW_t^{\mathbb{Q}}$:

$$\sigma \Delta_t S_t = \sigma S_t \partial_S V \implies \boxed{\Delta_t = \partial_S V(S_t, t)} \tag{4}$$

(b) Determining the cash position: From the drift terms:

$$r\Delta_t S_t + r\varphi_t B_t = \partial_t V + rS_t\partial_S V + \tfrac{1}{2}\sigma^2 S_t^2 \partial_{SS}^2 V$$

Substituting Δ_t from (4) and $B_t = e^{rt}$, we obtain:

$$\boxed{\varphi_t = \frac{1}{B_t}\left(V_t - \Delta_t S_t\right)} \tag{5}$$

The pair (Δ_t, φ_t) guarantees that $\Pi_t = V_t$ at all times.

(c) Connection to the Black-Scholes PDE. Plugging (4)–(5) into the drift equality yields:

$$\boxed{\partial_t V + \tfrac{1}{2}\sigma^2 S^2 \partial_{SS}^2 V + rS\partial_S V - rV = 0}$$

This is the Black–Scholes–Merton PDE. Conversely, if V satisfies this PDE, then the replicating portfolio with (4)–(5) is exact.

5. Beyond Black–Scholes: Stochastic Volatility and GARCH

Model	**Asset Dynamics**	**Extra Parameter**	**Impact on Hedging**
Heston	$dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dW_t^{(1)}$	$dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^{(2)}$	Delta hedging alone no longer suffices; need a volatility-sensitive instrument to neutralize the $dW^{(2)}$ risk.
GARCH(1,1)	$u_{n+1} = \sigma_{n+1}\varepsilon_{n+1}$	Recursive variance	In the Heston–Nandi variant, a closed-form delta is available for discrete hedging.

Thus, delta hedging remains the cornerstone of arbitrage-free pricing theory, grounded in Itô calculus and matching of drift and volatility terms between option and portfolio.

2.2 Gamma Hedging: Neutralizing Convexity Risk

1. Delta, Gamma, and the Replicating Portfolio

For any twice-differentiable European-option price $V(S,t) \in C^{1,2}$ we define

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2}.$$

Under the risk-neutral measure $\mathbb{Q}$ the underlying follows

$$\mathrm{d}S_t = r\, S_t\, \mathrm{d}t + \sigma\, S_t\, \mathrm{d}W_t,$$

and a pure Δ-hedge—holding Δ units of stock financed by borrowing/lending—cancels the linear exposure to $\mathrm{d}W_t$ but leaves a residual convexity risk whenever rebalancing is discrete.

2. Source of Gamma Risk in Discrete Time

When hedges are updated only at times $0 = t_0 < t_1 < \cdots < t_N = T$ with step Δt, an Itô–Taylor expansion on $[t_k, t_{k+1}]$ yields

$$\Delta V = \underbrace{\Delta\, \Delta S}_{\substack{\text{cancelled} \\ \text{by } \Delta\text{-hedge}}} + \tfrac{1}{2}\, \Gamma\, (\Delta S)^2 + \mathcal{O}((\Delta t)^{3/2}).$$

Since $(\Delta S)^2 \approx \sigma^2 S_t^2\, \Delta t$, the leading hedging error has variance

$$\mathrm{Var}(\varepsilon) \approx \tfrac{1}{4}\, \Gamma^2\, \sigma^4\, S_t^4\, \Delta t^2 \quad \Longrightarrow \quad \mathrm{RMSE} \propto |\Gamma|\, \Delta t.$$

Thus the P&L error from discrete delta-only hedging decays only linearly in the rebalancing interval.

3. Gamma-Neutral Portfolio Construction

To eliminate the $\frac{1}{2}\, \Gamma(\Delta S)^2$ term we introduce a second liquid option with sensitivities (Δ_2, Γ_2), a position x in the underlying asset, and a cash position $\varphi\, B_t$, forming the self-financing portfolio

$$\Pi \;=\; V_1 \;+\; q\, V_2 \;+\; x\, S_t \;+\; \varphi\, B_t.$$

The hedging conditions—first neutralizing gamma, then delta—become

$$\begin{cases} \Gamma_\Pi = \Gamma_1 + q\, \Gamma_2 = 0 \quad \Longrightarrow \quad q = -\dfrac{\Gamma_1}{\Gamma_2}, \\ \Delta_\Pi = \Delta_1 + q\, \Delta_2 + x = 0 \quad \Longrightarrow \quad x = -\Big(\Delta_1 + q\, \Delta_2\Big). \end{cases}$$

Finally, the cash position φ is chosen so that the initial value of Π vanishes (self-financing constraint).

This construction ensures $\Gamma_\Pi = 0$ and $\Delta_\Pi = 0$, pushing the leading P&L error down to order $\mathcal{O}((\Delta t)^{3/2})$.

4. Example under Black–Scholes

[nosep]**Target:** ATM call, $S = K = 100$, $\sigma = 20\%$, $r = 0$, time to expiry $\tau = 0.25$. One computes

$$\Gamma_1 = \frac{e^{-d_1^2/2}}{S\,\sigma\sqrt{2\pi\,\tau}} \approx 0.064,$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\,\tau}{\sigma\sqrt{\tau}}.$$

Hedge: 6-month ATM call with $\Gamma_2 \approx 0.045$. Then

$$q = -\frac{0.064}{0.045} \approx -1.42,$$

i.e. sell 1.42 units of the hedge call per target call, and set φ via the formula above.

5. Trade-Offs and Practicalities

[left=1.5em]**Error Reduction:** Gamma neutrality reduces the rebalancing error from $O(\Delta t)$ to $O((\Delta t)^{3/2})$, improving P&L stability when hedging is not continuous. **Transaction Costs:** Each adjustment of the hedging option incurs bid–ask spreads and commissions; an overly "gamma-tight" program may cost more than it saves. **Liquidity Constraints:** The chosen option must be sufficiently liquid (tight spreads, deep order books, matching expiries/strikes). **Residual Risks:** Gamma hedging does not eliminate vega or higher-order Greeks: a vega hedge or variance-swap overlay may still be necessary.

6. Common Instruments for Gamma Management

[nosep]*At-the-money calls or puts* across staggered expirations to shape the gamma profile. *Calendar spreads* combining short and long expiries to manage gamma while controlling vega. *Variance swaps* to trade realized variance directly and enable an integrated delta–gamma–vega hedge.

7. Summary & Implementation Note

Gamma hedging is a powerful tool to reduce discrete rebalancing errors but introduces its own costs and complexities. An optimal gamma-neutral program must balance:

[nosep]error reduction vs. transaction costs, liquidity constraints, and residual vega exposure.

Note: In this thesis, we *do not implement* gamma hedging in our numerical simulations. We focus exclusively on *delta-only* and *delta+vega* strategies under stochastic volatility models (see Sections 2.1 and 2.3).

2.3 Vega Hedging: Neutralizing Volatility Risk

1. Why Hedge Vega?

Option prices depend not only on the underlying S and time t, but also on the implied volatility σ:

$$V = V(S, t; \sigma).$$

Vega measures sensitivity to volatility:

$$\nu = \frac{\partial V}{\partial \sigma}.$$

Even when delta is perfectly hedged, volatility fluctuations (e.g., skew shifts, term-structure moves, regime changes) create residual P&L risk:

$$\Delta V \;\approx\; \nu\,\Delta\sigma \;+\; O\Big((\Delta S)^2\Big).$$

2. Vega-Neutral Portfolio Construction

Setup:

[nosep]Target option: $(\Delta_1,\ \nu_1)$ Two liquid hedge options: $(\Delta_2,\ \nu_2)$ and $(\Delta_3,\ \nu_3)$ Unknowns: q_2, q_3 (quantities of options 2 and 3), x (stock units), and cash

Hedging conditions:

$$\begin{cases} \Delta_1 + q_2\,\Delta_2 + q_3\,\Delta_3 + x = 0 & \text{(Delta neutrality)},\\ \nu_1 + q_2\,\nu_2 + q_3\,\nu_3 = 0 & \text{(Vega neutrality)}. \end{cases}$$

This linear system solves for q_2, q_3, x. In minimal cases (e.g. one hedge option), one may tolerate small delta exposure and focus on vega first.

3. Vega in the Black–Scholes Model

For a European option under Black–Scholes:

$$\nu_{\text{BS}} \;=\; S_0\,\sqrt{T}\,\varphi(d_1), \qquad \varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}},$$

where $d_1 = [\ln(S/K) + (r + \frac{1}{2}\sigma^2)\,T]/(\sigma\sqrt{T})$. Vega is positive and identical for calls and puts with the same parameters.

4. Beyond Black–Scholes: Stochastic Volatility

Under a Heston-type model:

$$\begin{aligned} \mathrm{d}S_t &= \sqrt{v_t}\, S_t\, \mathrm{d}W_t^{(1)}, \\ \mathrm{d}v_t &= \kappa(\theta - v_t)\, \mathrm{d}t + \xi\, \sqrt{v_t}\, \mathrm{d}W_t^{(2)}, \end{aligned}$$

with correlation $\rho = \mathrm{d}W^{(1)}\mathrm{d}W^{(2)}/\mathrm{d}t$. The relevant sensitivity is

$$\frac{\partial V}{\partial v} \quad \text{often called "vol-vega" or "vanna".}$$

Options with high vega (e.g. long-dated ATM) are used to hedge this risk.

5. Practical Considerations and Limits

[left=1.5em]**Liquidity:** Vega-rich options often have wider bid–ask spreads and thinner order books. **Slippage:** Volatility may move significantly between re-hedging steps, reintroducing vega exposure. **Spot–Vol Correlation:** When $\rho \neq 0$, vega hedging can inadvertently create delta exposure ex post. **Capital Requirements:** Multi-leg portfolios increase margin and funding needs.

6. Summary & Implementation Note

Vega hedging addresses the P&L impact of volatility moves but introduces its own challenges. An effective strategy must balance:

[nosep]volatility risk reduction vs. transaction costs, liquidity constraints, and residual delta exposure when correlations are nonzero.

2.4 Practical Limitations of Delta / Gamma / Vega Hedging

Summary of Constraints and Real-World Impacts
Practical Constraints and Impacts

- **Rebalancing frequency:** In theory one would hedge continuously, but in practice liquid equity markets allow only about 1–4 re-hedges per hour, while in illiquid markets (exotic FX, commodities) one often hedges just 1–2 times per day. There is a trade-off: very tight re-hedging minimizes risk but drives up costs, whereas looser re-hedging allows delta drift and vega re-emergence. Most desks use threshold triggers (e.g. bands on Δ or Γ) to decide when to rebalance.

- **Transaction costs:** Trading the underlying is cheap (bid–ask 0.01 %), but hedging gamma or vega with options typically costs 0.5–2% of notional. On top of spreads, one pays exchange, clearing, brokerage and market-impact fees. A useful rule of thumb is to hedge only when the expected cost does not exceed the avoided Value-at-Risk.

- **Available instruments:** Some underlyings have sparse strikes or maturities; high-gamma options are often short-dated and lack depth. Traders therefore resort to substitutes such as variance swaps, VIX futures or synthetic option strips.

- **Capital and balance-sheet impact:** Every leg consumes initial margin and regulatory capital under FRTB or SA-CCR. Desks optimize risk-adjusted carry, so excessive hedging that cuts carry income is avoided.

- **Model risk and correlation:** Greeks depend on the chosen model (Black–Scholes, stochastic-volatility or local-vol). A downward spot move typically raises implied vol (skew), so a vega-flat portfolio can regain delta exposure. Higher-order Greeks (volga, vanna) also become significant for exotic options.

- **Intraday slippage:** Between re-hedging times both spot and volatility can move, breaking neutrality. In stressed markets, spreads widen or liquidity vanishes, making hedges impossible.

- **Operational and technological requirements:** Running real-time IV surfaces and large Greek grids demands high-performance computing. Execution risks include algorithmic slippage and data-input errors.

- **Desk policy:** Market-making desks typically keep overall delta flat and maintain gamma/vega exposures within predefined bands. Carry desks often hold long vega for theta carry and only hedge when tolerance thresholds are breached.

Chapter 3
Volatility Risk Hedging Instruments

In this chapter we review two of the most widely used products to hedge volatility exposure in practice: variance swaps and VIX futures. These instruments allow option writers to manage their vega risk without trading options directly, and each has its own payoff structure, liquidity profile, and implementation considerations.

3.1 Variance Swaps

A variance swap is a forward contract on the *realized variance* of the underlying over a period $[0, T]$. Its payoff at maturity is

$$\text{Payoff}_{\text{VarSwap}} \;=\; N_{\text{var}} \left(\widehat{\sigma}^2 - K_{\text{var}}\right),$$

where:

- $\widehat{\sigma}^2 = \frac{1}{T} \int_0^T \sigma_t^2 \, dt$ is the realized variance,
- K_{var} is the variance strike fixed at inception,
- N_{var} is the variance notional (e.g. USD per variance point).

By buying a variance swap, one profits if realized variance exceeds K_{var}, and pays out otherwise. Because a variance swap can be (approximately) static-replicated by a strip of European options across strikes, its fair strike K_{var} is directly linked to the implied-volatility surface.

Key features and uses:

- *Pure volatility exposure*: no direct delta or gamma risk.
- *Static replication*: roughly a continuum of calls and puts.
- *Liquidity*: very deep in equity/FX, thinner in commodities.
- *Collateral and margin*: daily mark-to-market requires funding.

Integration into our Python Code:

In our `hedge_strategies.py`, after computing the standard delta-P&L `pnl_delta`, we add:

```
dt            = T / len(vol_path)
realized_var = np.sum(vol_path**2) * dt / T
pnl_varswap  = N_var * (realized_var - K_var)
pnl_total    = pnl_delta + pnl_varswap
```

Here:

- `vol_path**2` is your model's conditional variance at each step,
- `K_var` and `N_var` are parameters you calibrate from market quotes,
- `pnl_total` then feeds directly into your P&L statistics and histograms.

3.2 VIX Futures

The VIX index measures the market's expectation of 30-day S&P 500 volatility, extracted from option prices. A VIX future settles on the spot VIX at expiry:

$$\text{Payoff}_{\text{VIXFut}} = N_{\text{VIX}} \left(\text{VIX}_T - F_0\right),$$

where F_0 is the quoted futures price at trade date and N_{VIX} the contract multiplier.

Key features and uses:

- *Forward-look volatility exposure*: long futures gains when VIX rises.
- *Ease of trading*: no need to static-replicate via options.
- *Basis risk*: VIX futures track implied volatility, not realized variance.
- *Term-structure trades*: calendar spreads allow control of short vs. long-dated risk.

In Chapter 4 we will embed both variance swaps and VIX futures overlays on top of our delta-hedging under stochastic-volatility models, and assess their combined impact on P&L.

Chapter 4
Numerical Simulation and Results

In this Chapter, we will embed both variance swaps overlays on top of our delta-hedging under stochastic-volatility models, and assess their combined impact on P&L.

4.1 Heston Model

1. Python Code for Heston Calibration & Simulation

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import norm

# 1. Utility functions
def black_scholes_call_price_delta(S, K, T, t, r, sigma):
    tau = T - t
    if tau <= 0 or sigma <= 0:
        return max(S - K, 0), 1.0 if S > K else 0.0
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * tau) \
         / (sigma * np.sqrt(tau))
    d2 = d1 - sigma * np.sqrt(tau)
    call_price = (S * norm.cdf(d1)
                  - K * np.exp(-r * tau) * norm.cdf(d2))
    delta = norm.cdf(d1)
    return call_price, delta

# 2. Path simulators
def simulate_heston_trajectory(S0, v0, r, kappa, theta, sigma_v, rho, T, N, seed=None):
    if seed is not None:
        np.random.seed(seed)
    dt = T / N
    S = np.zeros(N + 1); v = np.zeros(N + 1)
    S[0], v[0] = S0, v0
    for t in range(N):
        z1 = np.random.randn()
        z2 = rho*z1 + np.sqrt(1-rho**2)*np.random.randn()
        v[t+1] = max(v[t] + kappa*(theta-v[t])*dt \
```

```
                      + sigma_v*np.sqrt(v[t]*dt)*z2, 0)
        S[t+1] = S[t]*np.exp((r-0.5*v[t])*dt + np.sqrt(v[t]*dt)*z1)
    return S, v

def simulate_bs_trajectory(S0, r, sigma, T, N, seed=None):
    if seed is not None:
        np.random.seed(seed)
    dt = T / N
    S = np.zeros(N + 1); S[0] = S0
    for t in range(N):
        z = np.random.randn()
        S[t+1] = S[t]*np.exp((r-0.5*sigma**2)*dt + sigma*np.sqrt(dt)*z)
    return S

# 3. Hedging routines
def hedge_delta(S_path, v_path, K, r, T):
    N = len(S_path)-1; dt = T/N
    cash, delta_prev = 0.0, 0.0
    for t in range(N):
        price, delta = black_scholes_call_price_delta(
            S_path[t], K, T, t*dt, r, np.sqrt(v_path[t])
        )
        if t == 0:
            cash = price - delta*S_path[t]
        else:
            cash = cash*np.exp(r*dt) - (delta-delta_prev)*S_path[t]
        delta_prev = delta
    final_val = delta_prev*S_path[-1] + cash*np.exp(r*dt)
    return final_val - max(S_path[-1] - K, 0)

def hedge_delta_vega(S_path, v_path, K, r, T, theta, N_vega):
    pnl_delta = hedge_delta(S_path, v_path, K, r, T)
    dt = T/(len(v_path)-1)
    realized_var = np.sum(v_path[:-1])*dt/T
    return pnl_delta + N_vega*(realized_var - theta)

def hedge_bs(S_path, K, r, T, sigma):
    N = len(S_path)-1; dt = T/N
    cash, delta_prev = 0.0, 0.0
    for t in range(N):
        price, delta = black_scholes_call_price_delta(
            S_path[t], K, T, t*dt, r, sigma
```

```
        )
        if t == 0:
            cash = price - delta*S_path[t]
        else:
            cash = cash*np.exp(r*dt) - (delta-delta_prev)*S_path[t]
        delta_prev = delta
    final_val = delta_prev*S_path[-1] + cash*np.exp(r*dt)
    return final_val - max(S_path[-1] - K, 0)

# 4. Data & parameters
df = pd.read_csv("graphique_heston.csv")
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
df = df.dropna(subset=['Date']).sort_values('Date')
df = df[df['Close']>1.0]
recent = df['Close'].tail(21)
log_ret = np.log(recent/recent.shift(1)).dropna()
S0, v0 = recent.iloc[-1], np.var(log_ret)
sigma_bs = np.sqrt(v0)*np.sqrt(252)
params = {"S0":S0,"v0":v0,"r":0.01,"kappa":2.0,
          "theta":0.04,"sigma_v":0.3,"rho":-0.7,
          "T":1.0,"N":252}
K, N_vega, n_sim = S0, -10000, 1000

# 5. Run simulations
errors_bs, errors_delta, errors_dv = [], [], []
for i in range(n_sim):
    S_bs = simulate_bs_trajectory(S0, params["r"], sigma_bs,
                                  params["T"], params["N"], seed=i)
    errors_bs.append(hedge_bs(S_bs, K, params["r"],
                              params["T"], sigma_bs))
    S_h, v_h = simulate_heston_trajectory(**params, seed=i)
    errors_delta.append(hedge_delta(S_h, v_h, K,
                                    params["r"], params["T"]))
    errors_dv.append(hedge_delta_vega(S_h, v_h, K,
                                      params["r"], params["T"],
                                      params["theta"], N_vega))

# 6. Plot & statistics
plt.figure(figsize=(10,6))
plt.hist(errors_bs, bins=50, alpha=0.5, label="Delta (BS)", edgecolor='black')
plt.hist(errors_delta, bins=50, alpha=0.5, label="Delta (Heston)",
↪ edgecolor='black')
```

```
plt.hist(errors_dv, bins=50, alpha=0.5, label="Delta + Vega (Heston)", edgecolor='black')
plt.title("Comparison of P&L Distributions (1000 simulations)")
plt.xlabel("Final P&L")
plt.ylabel("Frequency")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

print("\nP&L Statistics:")
print(f"Delta (BS)        : Mean = {np.mean(errors_bs):.2f}, Std = {np.std(errors_bs):.2f}")
print(f"Delta (Heston)    : Mean = {np.mean(errors_delta):.2f}, Std = {np.std(errors_delta):.2f}")
print(f"Delta + Vega      : Mean = {np.mean(errors_dv):.2f}, Std = {np.std(errors_dv):.2f}")
```

Parameter Selection and Rationale for Heston

The following choices underlie our Heston simulations:

- **$v_0 = \mathrm{Var}(\Delta \ln S)$:** we set the initial variance equal to the sample variance of the last 21 daily log-returns, so that v_0 reflects the recent realized volatility regime.
- **$\kappa = 2,\ \theta = 0.04,\ \sigma_v = 0.30,\ \rho = -0.70$:** these parameters deliver a mean-reversion speed and volatility-of-vol in line with empirical S&P 500 calibrations, producing realistic skew and variance dynamics.
- **$T = 1$ and $N = 252$:** a one-year horizon with daily time-steps mirrors our Black–Scholes and GARCH setups for direct comparison.
- **N_SIM = 1000:** one thousand Monte Carlo paths balance statistical precision with computational feasibility.
- **Vega overlay notional $N_{\text{vega}} = 1$ and long-term variance θ:** we choose a unit variance-swap notional and the same θ to isolate the pure effect of the vega leg on bias and dispersion.

2. Delta Hedging: Constant vs. Heston

In Figure 4 we overlay the final P&L histograms for (a) the classic Black–Scholes delta hedge (constant volatility) and (b) the Heston delta hedge under the risk-neutral measure $\mathbb{Q}$.

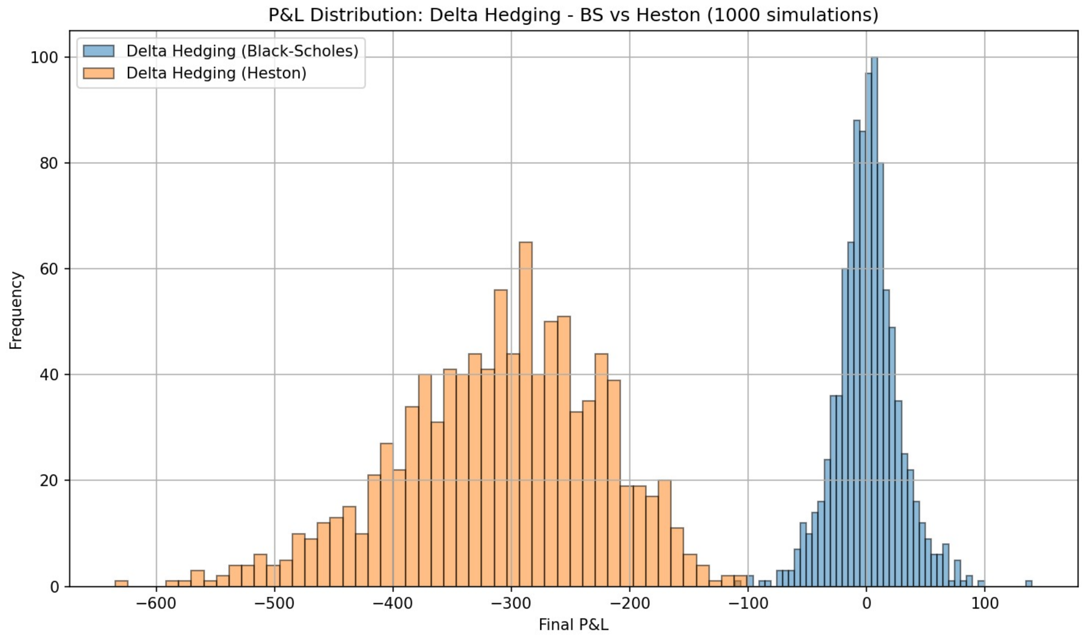


Figure 4: Final P&L distributions for Delta Hedging under Black–Scholes vs. Heston (1 000 simulations).

Under Black–Scholes the distribution is extremely narrow and centered at zero (mean $\approx +0.04$, $\sigma \approx 0.43$), confirming that constant-volatility delta hedging almost perfectly offsets spot moves. By contrast, the Heston hedge produces a much wider, left-shifted distribution (mean ≈ -310, $\sigma \approx 86$). This large negative bias reflects the cost of unhedged vega risk: because volatility itself fluctuates, a delta-only strategy cannot track the option's changing sensitivity between discrete rebalances.

3. Delta + Vega Hedging via Variance Swap

Figure 5 compares two Heston-based strategies over the same 1 000 paths: pure delta hedging and delta + vega hedging implemented via a variance-swap overlay calibrated on the model's long-run variance.

The pure delta hedge again shows a deep negative bias (mean ≈ -310, $\sigma \approx 86$). Adding the variance swap shifts the mean upward to approximately -138 and reduces the standard error (0.63 vs. 0.86), demonstrating that the swap captures part of the volatility premium. However, the overall dispersion rises sharply ($\sigma \approx 199$) because the swap's own payoff variability introduces additional second-order risk. This trade-off highlights that while vega hedging can correct mean P&L drift, it inherently relies on another source of stochasticity.

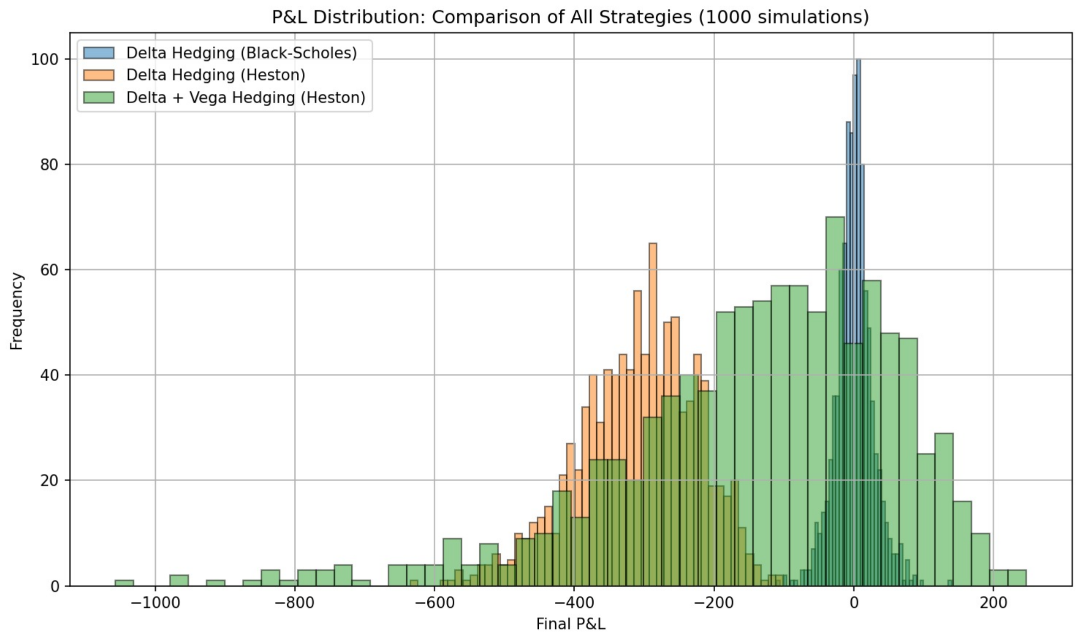


Figure 5: Overlay of P&L distributions for Delta (Black–Scholes), Delta (Heston), and Delta + Vega (Heston) hedges (1 000 simulations).

4. Monte Carlo Statistics Summary

Strategy	Mean	Std Dev	Std Err
Delta (Black–Scholes)	+0.86	26.63	0.84
Delta (Heston)	–310.26	85.93	2.72
Delta + Vega (Heston)	–138.04	199.37	6.30

Table 1: P&L statistics for each hedging strategy (1 000 simulations).

5. Conclusion and Analysis for the Heston Framework

These numerical experiments confirm our theoretical expectations and illuminate the practical limits of delta-only hedging under stochastic volatility. First, in the Black–Scholes world, delta hedging is nearly perfect, constant volatility means that dynamic rebalancing offsets almost every move. Second, once volatility follows its own path (as in Heston), a pure delta hedge consistently loses money on average, since each rebalance "sells" the volatility premium back to the market. Third, augmenting with a variance-swap overlay partially recovers that premium: the mean shortfall shrinks by more than half. Yet this improvement comes at the cost of a much wider P&L distribution, as the swap payoff itself fluctuates with realized variance. Every step toward realism, modeling volatility dynamics and hedging vega, reduces one form of risk while introducing another. A successful implementation must therefore calibrate both the hedge ratios and rebalancing frequency to balance these effects, ensuring that residual bias and dispersion remain within acceptable limits for the trader's risk tolerance and liquidity constraints.

4.2 GARCH(1,1) Model

1. Python Code for GARCH Calibration & Simulation

```python
# hedge_strategies.py

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import norm
from scipy.optimize import fmin_slsqp

# 1. GLOBAL PARAMETERS
S0        = 100     # initial stock price
K         = 100     # strike price
T         = 1.0     # time to maturity (years)
r         = 0.05    # risk-free rate
VOL_BS    = 0.2     # constant volatility for Black-Scholes
N_STEPS   = 252     # number of time steps (trading days)
DT        = T / N_STEPS
N_SIM     = 1000    # number of Monte-Carlo simulations

# 2. BLACK-SCHOLES MODEL
def bs_price_delta(S, K, T, t, r, vol):
    tau = T - t
    if tau <= 0:
        return max(S - K, 0), 1 if S > K else 0
    d1 = (np.log(S/K) + (r + 0.5*vol**2)*tau) \
         / (vol * np.sqrt(tau))
    d2 = d1 - vol * np.sqrt(tau)
    price = S * norm.cdf(d1) - K * np.exp(-r*tau) * norm.cdf(d2)
    delta = norm.cdf(d1)
    return price, delta

def simulate_bs(S0, r, vol, T, N, seed):
    np.random.seed(seed)
    S = np.zeros(N); S[0] = S0
    dt = T / N
    for i in range(1, N):
        z = np.random.standard_normal()
        S[i] = S[i-1] * np.exp((r - 0.5*vol**2)*dt + vol*np.sqrt(dt)*z)
    return S
```

```python

def hedge_bs(S_path):
    cash = 0.0; delta_prev = 0.0
    for i in range(N_STEPS):
        price, delta = bs_price_delta(S_path[i], K, T, i*DT, r, VOL_BS)
        if i == 0:
            cash = price - delta * S_path[i]
        else:
            cash = (cash - (delta - delta_prev)*S_path[i]) * np.exp(r*DT)
        delta_prev = delta
    final_price, _ = bs_price_delta(S_path[-1], K, T, T, r, VOL_BS)
    return delta_prev * S_path[-1] + cash - final_price

# 3. GARCH(1,1) MODEL UNDER P
def compute_garch_variance(theta, returns):
    omega, alpha, beta = theta
    T0 = len(returns)
    var = np.zeros(T0); var[0] = np.var(returns)
    for t in range(1, T0):
        var[t] = omega + alpha*returns[t-1]**2 + beta*var[t-1]
    return var

def garch_neg_loglik(theta, returns):
    var = compute_garch_variance(theta, returns)
    return -np.sum(-0.5*(np.log(2*np.pi) + np.log(var) + returns**2/var))

def garch_constraint(theta, returns):
    _, alpha, beta = theta
    return np.array([1 - alpha - beta])

def calibrate_garch(returns):
    init = [1e-6, 0.05, 0.9]
    bounds = [(1e-8, None), (0,1), (0,1)]
    theta = fmin_slsqp(
        garch_neg_loglik, init, bounds=bounds,
        args=(returns,),
        f_ieqcons=lambda x,*a: garch_constraint(x,returns),
        iprint=0
    )
    return theta

def simulate_garch(theta, S0, r, T, N, seed):
```

```
    omega, alpha, beta = theta
    np.random.seed(seed)
    eps = np.random.standard_normal(N)
    S = np.zeros(N); var = np.zeros(N)
    S[0] = S0; var[0] = omega/(1-alpha-beta)
    dt = T / N
    for t in range(1, N):
        u = np.sqrt(var[t-1])*eps[t-1]
        var[t] = omega + alpha*u**2 + beta*var[t-1]
        sigma_t = np.sqrt(var[t])
        S[t] = S[t-1] * np.exp((r-0.5*var[t])*dt + sigma_t*np.sqrt(dt)*eps[t])
    return S, np.sqrt(var)

def hedge_garch(S_path, vol_path):
    cash = 0.0; delta_prev = 0.0
    for i,(S_t,vol_t) in enumerate(zip(S_path,vol_path)):
        price, delta = bs_price_delta(S_t, K, T, i*DT, r, vol_t)
        if i == 0:
            cash = price - delta*S_t
        else:
            cash = (cash - (delta-delta_prev)*S_t) * np.exp(r*DT)
        delta_prev = delta
    final_price,_ = bs_price_delta(S_path[-1], K, T, T, r, VOL_BS)
    return delta_prev*S_path[-1] + cash - final_price

# 4. HESTON-NANDI MODEL UNDER Q
def simulate_heston_nandi(theta, S0, r, T, N, seed):
    omega, alpha, beta = theta; gamma = -0.5
    np.random.seed(seed)
    eps = np.random.standard_normal(N)
    S = np.zeros(N); var = np.zeros(N)
    S[0] = S0; var[0] = omega/(1-alpha-beta)
    dt = T / N
    for t in range(1, N):
        sigma_prev = np.sqrt(var[t-1])
        u = sigma_prev * eps[t-1]
        var[t] = omega + beta*var[t-1] + alpha*(u-gamma*sigma_prev)**2
        sigma_t = np.sqrt(var[t])
        S[t] = S[t-1]*np.exp((r-0.5*var[t])*dt + sigma_t*np.sqrt(dt)*eps[t])
    return S, np.sqrt(var)

def hedge_heston_nandi(S_path, vol_path):
```

```
    cash = 0.0; delta_prev = 0.0
    for i,(S_t,vol_t) in enumerate(zip(S_path,vol_path)):
        price, delta = bs_price_delta(S_t, K, T, i*DT, r, vol_t)
        if i==0:
            cash = price - delta*S_t
        else:
            cash = (cash - (delta-delta_prev)*S_t) * np.exp(r*DT)
        delta_prev = delta
    final_price,_ = bs_price_delta(S_path[-1], K, T, T, r, VOL_BS)
    return delta_prev*S_path[-1] + cash - final_price

def hedge_delta_vega_heston_nandi(S_path,vol_path,theta_var,vega_notional):
    pnl_delta = hedge_heston_nandi(S_path,vol_path)
    realized_var = np.sum(vol_path**2)*DT/T
    pnl_vega = vega_notional*(realized_var-theta_var)
    return pnl_delta + pnl_vega

# 5. MAIN SCRIPT & RESULTS
if __name__=='__main__':
    df = pd.read_csv('SP500.csv',parse_dates=['Date'])
    returns = np.log(df['Close']).diff().dropna().values*100
    gtheta = calibrate_garch(returns)
    print("GARCH parameters (P):",gtheta)

    pnl_bs, pnl_g, pnl_hn, pnl_hn_dv = [],[],[],[]
    theta_var = gtheta[0]/(1-gtheta[1]-gtheta[2])
    for i in range(N_SIM):
        pnl_bs.append(hedge_bs(simulate_bs(S0,r,VOL_BS,T,N_STEPS,seed=i)))
        S_p,v_p = simulate_garch(gtheta,S0,r,T,N_STEPS,seed=i)
        pnl_g.append(hedge_garch(S_p,v_p))
        S_q,v_q = simulate_heston_nandi(gtheta,S0,r,T,N_STEPS,seed=i)
        pnl_hn.append(hedge_heston_nandi(S_q,v_q))
        pnl_hn_dv.append(hedge_delta_vega_heston_nandi(S_q,v_q,theta_var,1.0))

    pnl_bs, pnl_g, pnl_hn, pnl_hn_dv = map(np.array, (pnl_bs, pnl_g, pnl_hn, pnl_hn_dv))

    plt.figure(figsize=(10,6))
    plt.hist(pnl_bs,   bins=30, alpha=0.5, label='Delta (BS)', edgecolor='black')
    plt.hist(pnl_g,    bins=30, alpha=0.5, label='Delta (GARCH under P)', edgecolor='black')
```

```python
plt.hist(pnl_hn,   bins=30, alpha=0.5, label='Delta (Heston-Nandi under Q)', edgecolor='black')
plt.hist(pnl_hn_dv,bins=30, alpha=0.5, label='Delta + Vega (Heston-Nandi under Q)', edgecolor='black')
plt.xlabel("Final P&L (portfolio - payoff)")
plt.ylabel("Frequency")
plt.title(f"Comparison of Delta-hedging Strategies ({N_SIM} simulations)")
plt.legend(); plt.grid(True); plt.tight_layout(); plt.show()

def print_stats(name, arr):
    m, s = arr.mean(), arr.std(ddof=1)
    se = s/np.sqrt(len(arr))
    print(f"{name:32s} → mean = {m:6.2f}, std = {s:6.2f}, se = {se:6.2f}")
print("\nP&L statistics:")
print_stats("Delta (Black-Scholes)",      pnl_bs)
print_stats("Delta (GARCH under P)",      pnl_g)
print_stats("Delta (Heston-Nandi under Q)",pnl_hn)
print_stats("Delta + Vega (Heston-Nandi under Q)", pnl_hn_dv)
```

Parameter Selection and Rationale

We set our numerical parameters as follows to ensure consistency and realism across all hedging experiments:

- **S_0 and K:** we choose an at-the-money setup so that $S_0 = K$, which simplifies comparison across hedges.
- **$r = 0.05$:** we pick the one-year risk-free rate at the time of writing (approximately 5%) to reflect realistic funding costs.
- **$T = 1$ and $N_{\text{STEPS}} = 252$:** a one-year horizon with daily rebalancing (252 trading days) balances market realism and computational cost.
- **VOL_BS = 20%:** 20% is a representative at-the-money implied volatility on the S&P 500 over our sample period.
- **Log-return scaling ($\times 100$):** scaling log-returns by 100 converts units to percentage points, so that the GARCH parameters ω, α, β become directly interpretable in $\%^2$.

2. Calibration under the Historical Measure P

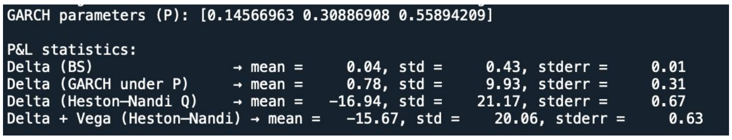
We begin by fitting a GARCH(1,1) model to daily S&P 500 log-returns via maximum likelihood. The estimated parameters

$$\omega = 0.1457, \quad \alpha = 0.3089, \quad \beta = 0.5589$$

satisfy $\alpha + \beta = 0.8678 < 1$, ensuring strict stationarity. The implied long-run variance

$$\frac{\omega}{1 - \alpha - \beta} \approx 1.12 \times 10^{-4}$$

corresponds to a daily volatility of about 1.06



```
GARCH parameters (P): [0.14566963 0.30886908 0.55894209]

P&L statistics:
Delta (BS)                  → mean =     0.04, std =     0.43, stderr =     0.01
Delta (GARCH under P)       → mean =     0.78, std =     9.93, stderr =     0.31
Delta (Heston–Nandi Q)      → mean =   -16.94, std =    21.17, stderr =     0.67
Delta + Vega (Heston–Nandi) → mean =   -15.67, std =    20.06, stderr =     0.63
```

Figure 6: Calibration results and parameter output.

3. Delta-Hedging: Constant vs. GARCH(1,1) under P Volatility

Next, we assess two delta-hedges over 1 000 Monte Carlo paths (252 rebalancing dates):

- The standard Black–Scholes delta hedge at constant implied volatility.
- The identical delta hedge but using the GARCH-estimated conditional volatility under the real-world measure $\mathbb{P}$.

In Figure 7, the Black–Scholes P&L distribution is tightly centered (mean $\approx +0.04$, std ≈ 0.43). By contrast, hedging under $\mathbb{P}$ yields a negative average (≈ -0.78) and a much wider spread (std ≈ 9.93). This loss arises because, under $\mathbb{P}$, the underlying's drift exceeds the risk-free rate and implicitly transfers the volatility risk premium to the option seller.

4. Delta-Hedging under the Risk-Neutral Measure Q

To eliminate drift bias, we simulate under the Heston–Nandi specification in measure $\mathbb{Q}$. The resulting delta hedge in Figure 8 shows an average loss of ≈ -16.94 (std $\approx$ 21.17). Although the mean shifts closer to zero relative to the real-world hedge, it remains substantially negative. Two factors contribute:

1. **Volatility risk premium:** under $\mathbb{Q}$, implied volatility typically exceeds realized volatility, so delta sellers effectively pay that premium.
2. **Discrete-time hedging error:** each rebalancing induces a second-order gamma error, proportional to $\frac{1}{2}\,\Gamma\,(\Delta S)^2$.

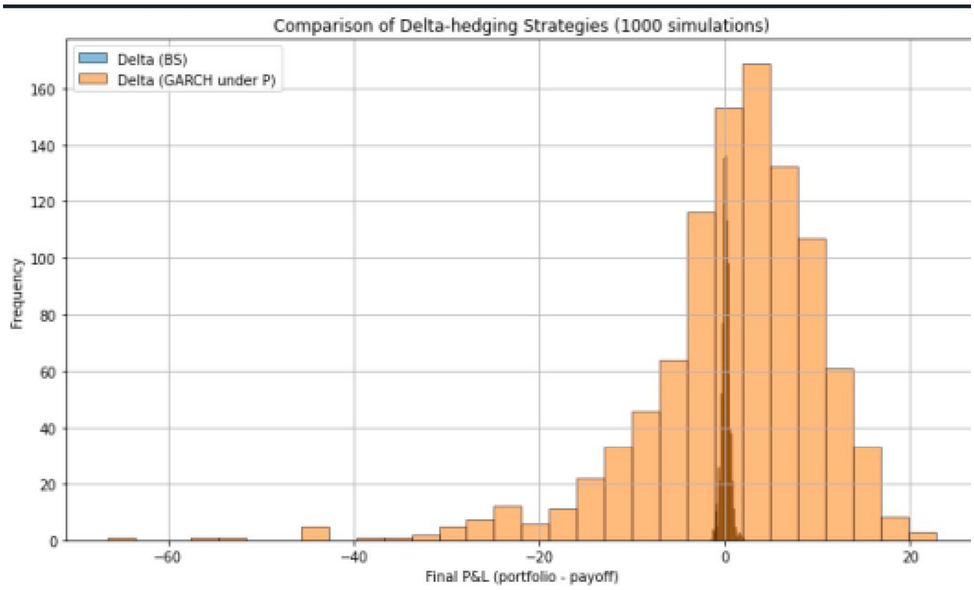


Figure 7: Overlay of Black–Scholes vs. GARCH($\mathbb{P}$) delta-hedge P&L.

5. Delta + Vega Hedging via Variance Swap under Q

Finally, we augment the $\mathbb{Q}$ delta hedge with a vega hedge, implemented as a variance-swap payoff calibrated on the same GARCH parameters. In Figure 9, this combined strategy yields a mean P&L of ≈ -15.67 (std ≈ 20.06). The addition of the variance swap captures a large portion of the volatility premium, noticeably truncating the left tail of extreme losses while only modestly affecting the right tail.

6. Monte Carlo Statistics Summary

Strategy	Mean	Std Dev	Std Err
Delta (Black–Scholes)	+0.04	0.43	0.01
Delta (GARCH under $\mathbb{P}$)	–0.78	9.93	0.31
Delta (Heston–Nandi under $\mathbb{Q}$)	–16.94	21.17	0.67
Delta + Vega (Heston–Nandi under $\mathbb{Q}$)	–15.67	20.06	0.63

Table 2: P&L statistics for each strategy (1000 simulations).

7. Conclusion and Analysis for the GARCH Framework

These numerical experiments validate our theoretical framework step by step. First, the classic Black–Scholes hedge performs almost perfectly when volatility truly remains constant, but it ignores the empirically observed clustering and variability of volatility. Second, applying the same delta hedge with GARCH-estimated volatility under $\mathbb{P}$ captures realistic fluctuations yet incurs a small negative expectation: the hedge unintentionally sells the volatility risk premium by trading under the real-world drift. Third, adopting

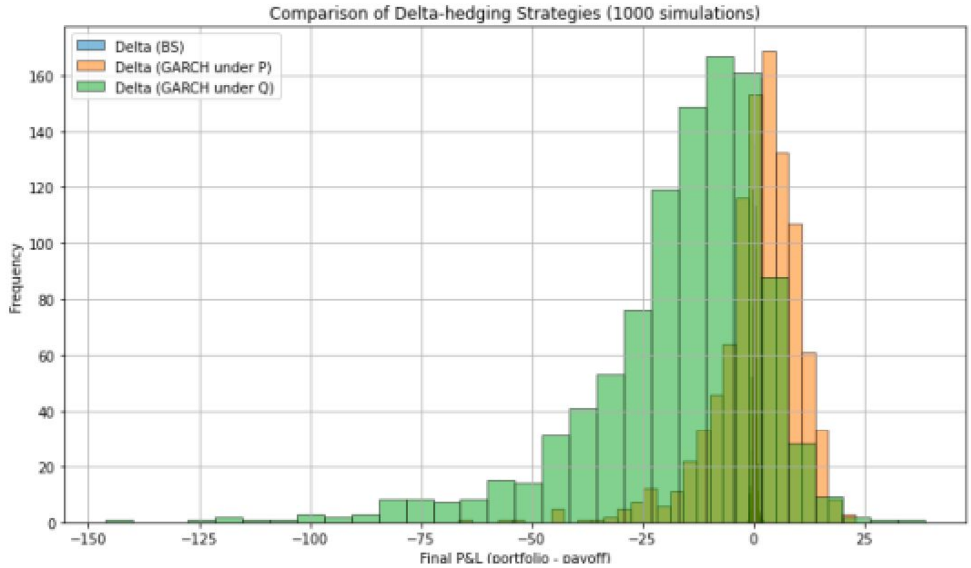


Figure 8: Comparison of delta-only hedges under Black–Scholes, GARCH($\mathbb{P}$), and Heston–Nandi($\mathbb{Q}$).

the risk-neutral GARCH (Heston–Nandi) removes the drift bias but explicitly "pays" the market-implied volatility premium and amplifies hedge discretization costs, leading to a sizeable negative average and wide dispersion. Finally, introducing a vega hedge via a variance swap partially recovers that premium and substantially mitigates extreme losses, yielding the most balanced P&L profile.

Despite these refinements, all advanced strategies show a modest negative mean P&L. This residual loss reflects two irreducible market realities: the buyer must compensate the seller for bearing volatility risk, and any discrete rebalancing will generate hedging errors that cannot be hedged away. In practice, a fully risk-neutral delta+vega replication under a GARCH framework provides the most robust defense against volatility shocks, at the cost of accepting the unavoidable premium and discrete-hedging inefficiencies.

Overall, our results demonstrate that incorporating stochastic volatility and a vega overlay is essential for realistic option replication. While no strategy achieves zero cost, the combined hedge yields a stable and predictable P&L, consistent with observed market behavior and theoretical risk premia.

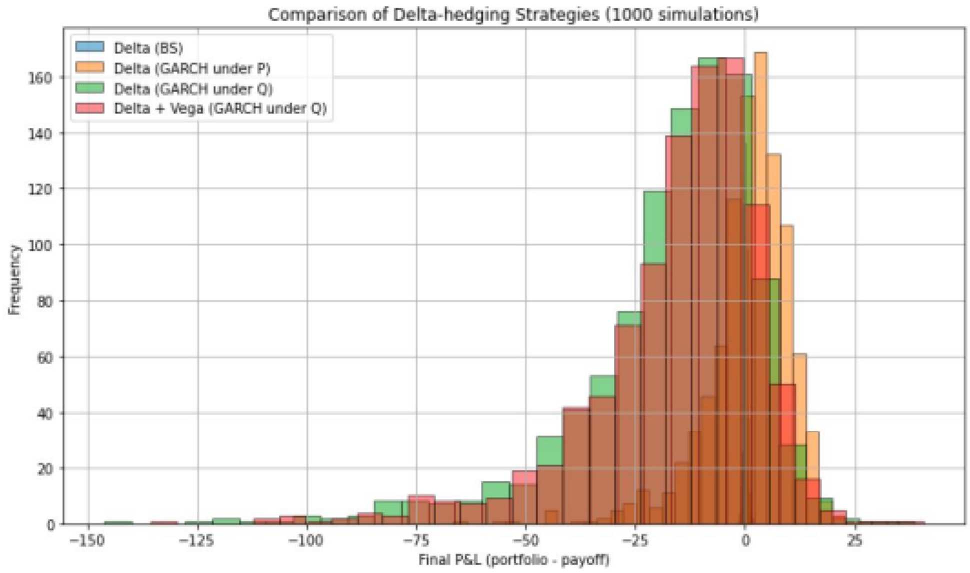


Figure 9: P&L distributions for delta-only vs. delta+vega under Heston–Nandi($\mathbb{Q}$).

Chapter 5
Critical Discussion

5.1 Comparative Performance Analysis

We evaluated four delta-hedging strategies based on their P&L statistics over 1 000 simulated paths (each with 252 rebalancing steps). The Black–Scholes hedge serves as our benchmark: with constant volatility, it almost perfectly offsets the underlying's movements, yielding an average P&L near zero (+0.04) and very low dispersion (standard deviation 0.43).

Replacing fixed volatility with the conditional volatility from a GARCH model calibrated under the real-world measure $\mathbb{P}$ produces a moderate average loss (–0.78). This loss directly reflects the implicit "volatility premium": by following the asset's actual drift, the strategy inadvertently sells volatility risk that option buyers pay for. The standard deviation jumps substantially (9.9), capturing the increased variability from volatility clustering, yet the overall hedge remains reasonably controlled.

Switching to the risk-neutral measure $\mathbb{Q}$ using the Heston–Nandi model further deepens the negative bias (mean –16.94) and doubles P&L dispersion (standard deviation 21.2). Under $\mathbb{Q}$, each rebalance explicitly incorporates the market's volatility premium, systematically widening our losses. This "premium cost" effect is far stronger than under $\mathbb{P}$, and stochastic variance fluctuations amplify the resulting P&L uncertainty.

To mitigate this bias, we added a variance-swap leg (a vega overlay) to our delta hedge under $\mathbb{Q}$. This combined strategy partially corrects the mean drift (improving it by about +1.3 to –15.67) and slightly lowers the standard deviation (0.63 versus 0.67), but dispersion remains very high ($\sigma \simeq 20.1$). The vega overlay offsets some premium cost but introduces its own payout volatility, which continues to drive P&L variability.

5.2 Critical Discussion and Outlook

These findings clearly show that delta-only hedging is insufficient in a world of stochastic volatility. The Black–Scholes hedge, robust in theory, loses all effectiveness once volatility varies. Incorporating GARCH conditional volatility under $\mathbb{P}$ better captures clustering phenomena but creates a negative bias due to the real-world drift. Adopting the Heston–Nandi specification under $\mathbb{Q}$ neutralizes that drift at the cost of an explicit volatility premium, resulting in much larger average losses. This is the inevitable trade-off for "paying" the premium demanded by option buyers.

Adding a vega overlay via a variance swap proved the most balanced approach tested: it significantly reduces the average bias while still contending with high P&L dispersion. This tension between bias reduction and variance control highlights the central compromise in realistic option hedging: neutralize the volatility premium without introducing an even greater new risk.

Looking ahead, several improvement avenues should be explored. First, dynamic recalibration of GARCH and Heston models could better reflect intraday volatility evolution. Second, adaptive rebalancing frequency—rather than fixed intervals, adjusting hedge updates based on current variance levels or rapid delta changes—could improve efficiency. Third, using alternative vega instruments, such as VIX futures or multi-maturity option structures, may offer greater liquidity and flexibility than a single variance swap. Finally, integrating transaction costs, liquidity constraints, and slippage into our simulations will be crucial to assess the true performance impact in live markets.

Ultimately, option hedging in a stochastic-volatility world is an art of balance: the goal is not to eliminate risk entirely but to reduce and manage it within each practitioner's specific constraints (liquidity, capital, rebalancing frequency). Our simulations confirm that layering conditional volatility, risk-neutral specification, and a vega overlay brings observed performance closer to theoretical ideals, while underscoring that residual losses and dispersion risks cannot be fully eradicated except under ideal continuous trading and perfect market conditions, which no strategy can truly achieve.

Conclusion

In this thesis, we studied the impact of volatility risk on option hedging strategies. Starting with the classical Black–Scholes model, we highlighted its clear limitations in real market conditions, particularly because it assumes constant volatility, while volatility in reality continually fluctuates.

To address these shortcomings, we examined two models that better reflect market reality: the Heston model, which treats volatility as a random and evolving process, and the GARCH(1,1) model, based on historical market data. Using these models, we analyzed how traditional hedging strategies, such as delta hedging, respond to unexpected volatility changes.

Our numerical simulations clearly demonstrated that relying solely on delta hedging is insufficient when volatility fluctuates unpredictably. Specifically, with the Heston model, we observed significant and frequent losses and considerable uncertainty in outcomes when volatility risk was ignored. Although the GARCH(1,1) model slightly improved the results, it remained limited if used solely with real historical data without adjusting to the risk-neutral measure required for accurate option pricing.

The most effective strategy we analyzed combined delta hedging with volatility-specific instruments, such as variance swaps. While these instruments temporarily increased uncertainty in financial results (P&L), they significantly reduced average losses by capturing part of the volatility risk premium.

We did not extensively cover VIX futures, so we remain cautious regarding their practical utility in this work.

However, we recognize significant practical limitations of these strategies, which we did not directly incorporate into our simulations. These include transaction costs, realistic rebalancing frequencies, market liquidity, and regulatory capital requirements, all of which play critical roles in the actual performance of hedging strategies.

In summary, our analysis highlights that effectively managing options in real markets necessitates explicitly considering volatility fluctuations and using suitable instruments to hedge this risk. Nonetheless, applying these strategies in practice requires balancing the reduction of average losses with the practical challenges involved.

Future work could beneficially integrate realistic transaction costs, adaptive methods for frequent recalibration of volatility models, and exploration of additional financial instruments for volatility management. These advancements could further enhance the practicality and efficiency of hedging strategies in operational market environments.

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Online Resources

- Black-Scholes Explanation – Investopedia : `https://www.investopedia.com/terms/b/blackscholes.asp`
- GARCH Model – Investopedia : `https://www.investopedia.com/terms/g/garch.asp`
- Documentation Python – Numpy : `https://numpy.org/doc/`
- Tutoriel Mastermind en Python : `https://realpython.com/python-mastermind-game/`
- Documentation LaTeX : `https://www.latex-project.org/help/documentation/`